

AORANE

•Y "Ù " 8"y9'BÂ 9'éG"i> 'YG ")?•ò R Your Health, In Your Hands

Complete Platform Blueprint & Product Analysis Document

Comprehensive analysis of all four product verticals —
Mobile App · Business Portal · Admin Console · Landing Page

4

Product Verticals

25+

Feature Modules

40+

API Endpoints

15+

DB Tables

Prepared for:

Mangesh Angadverma | Founder & CEO, AORANE Technologies

Version 1.0 · April 2025 · CONFIDENTIAL

Health-Tech

Made in India

AI-Powered

DPDPA Compliant

B2B + B2C

Table of Contents

01	Executive Summary	03
02	Platform Architecture Overview	04
03	Landing Page — aorane.com	05
04	Mobile App — AORANE Health	07
05	Business Portal — Aorane Business	11
06	Admin Console — Aorane Admin	14
07	Backend API Server	17
08	Database Schema	19
09	Tech Stack & Integrations	20
10	Subscription & Monetization	21
11	Security & Compliance	22
12	Deployment Architecture	23

01 Executive Summary

AORANE is an Indian Health-Tech platform that provides a unified ecosystem for personal health management, corporate wellness, and AI-powered medical intelligence. The platform serves both B2C (individual users and families) and B2B (corporates, hospitals, gyms, schools) segments through four tightly integrated product verticals.

Mission Statement

Core Mission

"Aachi Sehat, Hamesha ke Liye" — To make quality health intelligence accessible to every Indian citizen through AI-powered tracking, emergency services, and preventive wellness tools.

Product Verticals at a Glance

AORANE Mobile App
Personal health tracking, AI food scan, stress management, medicine reminders, blood donation, family health — the core consumer product targeting individuals and families across India.

AORANE Business Portal
B2B SaaS platform for corporates, hospitals, gyms & schools to monitor aggregate employee/member health, manage wellness programs, and generate compliance reports.

AORANE Admin Console
Internal superadmin panel for platform governance — user moderation, revenue analytics, AI model configuration, feature flags, blood emergency management.

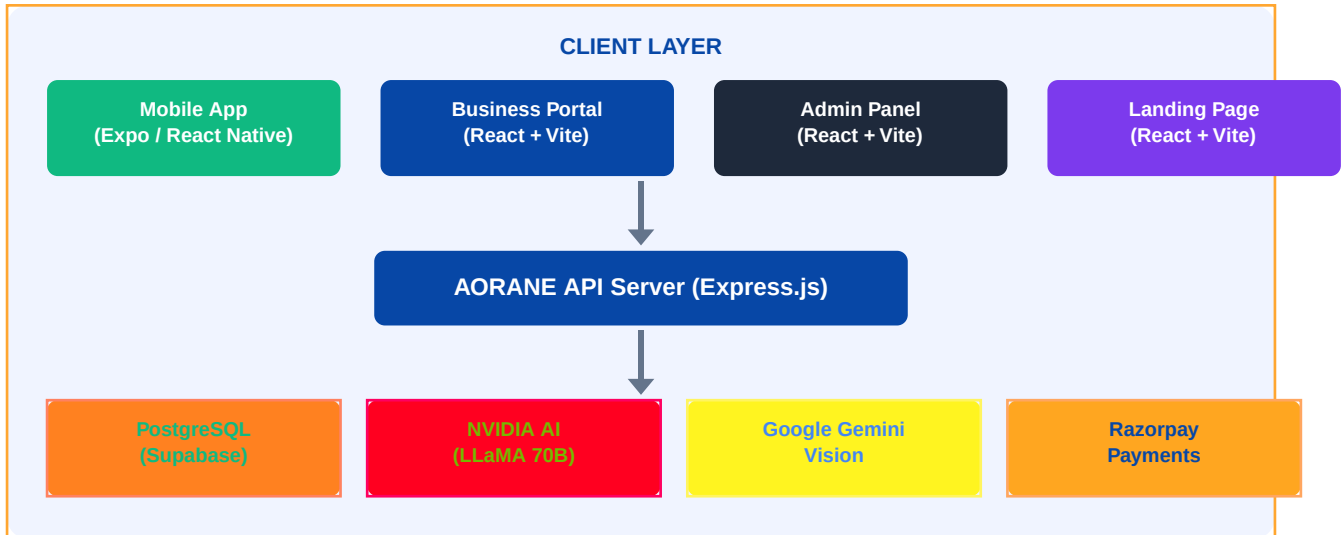
AORANE Landing Page
High-conversion marketing site (aorane.com) with dynamic B2C/B2B audience toggle, pricing, features showcase, and app download CTA.

Key Business Metrics (Platform Targets)

Metric	Current / Target
Registered Users	2L+ (200,000+)
Play Store Rating	4.8&
Business Partners	500+ Companies
India Regions	Pan-India (Singapore server)
Data Compliance	DPDP Act 2023, HIPAA-ready

02 Platform Architecture Overview

AORANE follows a monorepo architecture with 5 packages sharing a single PostgreSQL database (Supabase). All client apps communicate through a centralized Express.js API server.



Monorepo Package Structure

Package	Technology	Purpose
@workspace/aorane-mobile	Expo + React Native	Consumer mobile app (iOS/Android/Web)
@workspace/business-portal	React 18 + Vite + Tailwind	B2B organization management portal
@workspace/admin-panel	React 18 + Vite + Tailwind	Internal superadmin console
@workspace/aorane-landing	React 18 + Vite + Tailwind	Public marketing website
@workspace/api-server	Express.js + TypeScript	Centralized REST API + business logic
@workspace/db	Drizzle ORM + PostgreSQL	Shared database schema + migrations

03 Landing Page — aorane.com

Overview

A high-conversion, single-page React application serving as the primary marketing surface for AORANE. Features a dynamic "Audience Toggle" that re-skins the entire page between B2C (individual health) and B2B (enterprise wellness) messaging.

Page Sections & Structure

Hero Section	Dynamic headline + dashboard preview. Toggle: "For You" vs "For Business". CTA: "Download App" / "Get Started Free"
Stats Bar	Social proof: 2L+ Users, 4.8& Play Store, 500+ Companies trusted
Bento Grid	Visual showcase of key app screens — AI Food Scan, Health Score, Sleep Analytics
Features Grid	Icon-based showcase of 12+ technical capabilities with descriptions
How It Works	3-step onboarding flow illustration for new users
Testimonials	Rotating carousel of user/client feedback stories
Pricing Section	Interactive toggle: Monthly/Yearly × Individual/Business plans with Razorpay checkout
App Download	Full-width CTA section with Play Store badge and QR code
Trust Section	Security badges: 256-bit TLS, DPDPA Compliant, ISO 27001
Footer	Links to Privacy Policy, Terms of Service, Contact, Social Media

Key Features Highlighted (B2C)

- AI Food Scanner — Photo-based macro identification using Google Gemini Vision
- NVIDIA Health Intelligence — Personalized health predictions using LLaMA 3.3 70B
- MET Exercise Engine — Calorie tracking for 300+ exercise types
- Smart Medical Scan — OCR for prescriptions and lab reports
- Family Health Hub — Manage health for up to 4 family members
- WhatsApp Health Bot — (Coming Soon) Log meals via WhatsApp chat
- Blood Emergency Network — Real-time blood donor finder + emergency requests
- Period & Cycle Tracker — AI-powered menstrual health predictions

Key Features Highlighted (B2B)

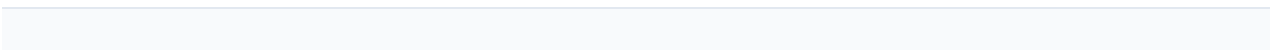
- Organization Portal — Manage 1000+ employees from a single dashboard
- Population Analytics — AI-powered risk stratification and burnout detection
- Real-time Monitoring — Department-wise aggregate health dashboards
- DPDPA Compliant — Individual data stays private, only aggregates shown to org admin
- GST-ready Invoicing — Automated invoice generation for Indian businesses
- Custom Wellness Programs — Step contests and health challenges

Routes

- / — Main landing (LandingPage.tsx)
- /features — Feature deep-dive (FeaturesPage.tsx)
- /pricing — Pricing plans (PricingPage.tsx)
- /business — B2B specific landing (BusinessPage.tsx)
- /privacy — Privacy Policy
- /terms — Terms of Service

Tech Stack — Landing Page

Layer	Technology
-------	------------



Vite

Wouter (lightweight)

Animations

Framer Motion

Charts

Recharts

Payments

04 Mobile App — AORANE Health

Overview

The core consumer product — a comprehensive AI-powered health and wellness app built with Expo (React Native). Runs on Android, iOS, and Web. Features 15+ health tracking modules, AI diagnostics, emergency services, and a clinically-informed stress tracker.

Navigation Architecture

The app uses Expo Router with a bottom tab bar navigation. The center tab features a prominent "AI Scan" quick-access button for the most-used feature.

- Tab 1 — Dashboard (Home): Health command center
- Tab 2 — Exercise (Fitness): Workout logging and calorie tracking
- Tab 3 — AI Scan (Center): Food scan, medical report scan, smart analysis
- Tab 4 — Medicine: Pill reminders and compliance tracking
- Tab 5 — Profile: Settings, subscription, notifications

Authentication & Security

Feature	Implementation
OTP Login	Mobile number !' SMS/WhatsApp OTP !' JWT tokens
Email Login	Email OTP verification
Google Sign-In	Firebase Google OAuth + ID token verification
PIN Lock	4-digit PIN set on first login, required on app re-open
Biometric	Placeholder implemented (TouchID/FaceID for future)
Auto Logout	Session expiry with PIN verification screen
Token Refresh	Silent JWT refresh using refresh token

Onboarding Flow (First-Time User)

- Step 1 — Physical Details: Height, weight, age, gender, blood group
- Step 2 — Lifestyle: Smoking/drinking habits, work profile, sleep hours
- Step 3 — Health History: Chronic conditions (Diabetes, BP, Thyroid, etc.)
- Step 4 — Goals: Weight loss, muscle gain, stress management, etc.
- Step 5 — Permissions: Camera (food scan), Location (blood donors), Notifications

04 Mobile App — Feature Modules (Continued)

Dashboard Screen

- AORANE Health Score card — daily score (0-100) with grade (A/B/C/D)
- Calorie Summary — eaten vs burned with macro breakdown (Protein/Carbs/Fat)
- Water Tracker — interactive dots UI, haptic feedback on each glass logged
- Stress Check-in Card — latest score + burnout risk alert + quick-log shortcut
- Medicine Preview — next 3 scheduled medicines for the day
- AI Feature Tiles — Daily Coach (nutrition AI) + Intelligence (health prediction)
- Quick Services Grid — 6 one-tap shortcuts to core features

Food & Nutrition Tracking

- Food Log — Breakfast/Lunch/Dinner/Snack sections with calorie totals
- AI Food Scan — Photo upload ! Gemini Vision identifies food ! calories + macros
- Text Search — 4-level lookup: History ! Internal DB ! AI Cache ! Live AI
- Food Favorites — Most logged items for quick re-entry
- Weather-based Suggestions — AI recommends food based on local weather/season
- Daily Coach — Personalized nutrition advice based on 30-day history
- Meal Swap — AI suggests healthier alternatives for logged meals
- Health Score per meal — 1-10 rating with explanation

Stress Tracker (Clinically-Informed)

- Five-Pillar Score — Real-time stress from Food + Sleep + Water + Exercise + Medicine data
- Full Clinical Assessment — PSS-inspired questionnaire with mood, energy, symptoms
- Quick Mood Log — Single tap mood entry (Happy/Sad/Nervous/Mood Off)
- Today / Weekly Summary — Today's avg, weekly avg with 7-day bar chart
- 30-Day Personal Baseline — Compares this week vs your personal historical avg
- Burnout Risk Detection — Auto-alert when score consistently above 70+
- 4-7-8 Breathing Exercise — Interactive animated breathing pacer for relaxation
- Work Profile Penalty — 16 work profiles mapped to stress modifier (e.g. surgeon = +15)

Medicine Module

- Medicine Schedules — Add medicine with name, dosage, frequency (daily/weekly/custom)
- Meal Timing — Before/After Breakfast, Lunch, Dinner, Bedtime
- Dose Tracking — Mark as Taken / Missed / Skipped with timestamp
- Compliance Score — Taken/Scheduled ratio used in health score + stress calculation
- Refill Alerts — Notifications when supply runs low (future)
- Dashboard Preview — Next 3 medicines shown on home screen

Blood Donation Network

- Donor Registration — Blood group, city, GPS coordinates, OTP verification
- Find Donors — Search by blood group + city, or GPS "Near Me" (50km radius)
- Emergency Request — Create urgent blood request (OTP verified, hospital info required)
- Emergency Direct — Simplified 2-tap emergency request for mobile
- Donation Confirmation — Records donation + automatic 90-day donor cooldown
- Admin Moderation — All requests reviewed by admin panel for authenticity

04 Mobile App — Feature Modules (Continued)

AI & Intelligence Features

- Health Prediction — 30-day data analysis !' disease risk scores (Diabetes, BP, Thyroid, etc.)
- Risk Levels — Low / Moderate / High / Critical with color-coded alerts
- Weekly Diet Chart — AI generates 7-day Indian meal plan (Breakfast/Lunch/Dinner/Snacks)
- Smart Medical Scan — Upload lab report !' AI extracts findings !' vs normal range analysis
- NVIDIA LLaMA 3.3 70B — Used for fast text-based nutrition and health queries
- Google Gemini Vision — Used for image-based food scan and medical report OCR
- AI Model Routing — Admin-configurable per feature (can switch models without code change)

Exercise Tracker

- Activity Logging — 300+ exercise types (Walking, Running, Yoga, Swimming, Gym, etc.)
- MET-Based Calculation — Scientific calorie burn estimation using Metabolic Equivalent
- Duration + Intensity — Tracks minutes and intensity level per session
- Weekly Activity Summary — 7-day consistency score and total calories
- Health Score Impact — Exercise directly feeds into daily AORANE Health Score

Water & Wellness Trackers

- Water Tracker — Glass-by-glass tracking, daily goal (default 8 glasses), % completion
- Sleep Tracking — Logs hours and quality rating, feeds into five-pillar stress score
- Period Tracker — Cycle logging (flow, symptoms), AI-predicted next date + fertile window
- Body Metrics — Manual entry for SpO2, heart rate, blood pressure, weight (from wearable/manual)

Wearable & Google Fit Integration

- Google Fit OAuth — One-time auth, syncs steps, calories, heart rate, sleep
- Manual Entry Mode — For non-supported bands: enter SpO2, distance, steps manually
- 7-Day Averages — Summary of all synced wearable metrics
- Auto Health Score Update — Wearable data counts toward daily health score calculation

Health Report & Scorecard

- PDF Health Report — Weekly/Monthly professional report (printable, shareable)
- Report Contents: BMI analysis, nutrition score, hydration, exercise, medicine compliance, stress trends, AI insights
- Health Scorecard — Virtual "AORANE Health ID Card" with Aorane ID, blood group, BMI, health grade
- Custom Avatar — User can set profile picture on the health card
- Share/Download — Report can be shared as PDF with doctors or family

Emergency Services

- Blood Emergency — One-tap emergency blood request with hospital info
- Medical Emergency Hub — Quick dial for 112, 108, 100, 101 national emergency numbers
- Emergency Contacts — Personal contacts alerted during SOS
- Accident Emergency — (Coming Soon) 3-tap system to alert nearest hospitals with GPS

Family Health Hub

- Create Group — Generate FAMxxxx invite code, add up to 4 family members
- Family Plan — Shared subscription for the whole family
- Health Overview — See all family members' latest health scores in one view
- Cross-member Monitoring — Ideal for elderly parents or children's health tracking

05 Business Portal — Aorane Business

Overview

A B2B SaaS CRM platform for organizations to monitor aggregate employee/member health. Built with React + Vite + Tailwind. Fully DPDPA Act 2023 compliant — individual sensitive data is never exposed to org admins, only privacy-safe aggregate metrics.

Target Organizations

Organization Type	Primary Use Case	Key Value
Corporates & Enterprises	Employee wellness monitoring	Reduce absenteeism by 31%
Hospitals & Clinics	Staff fitness + shift readiness	Monitor clinical staff health
Gyms & Fitness Centers	Member progress tracking	Improve member retention
Yoga & Wellness Studios	Stress score monitoring	Mindfulness trend analysis
Schools & Colleges	Student + staff health cards	Annual health report cards
Insurance & TPAs	Risk assessment	Verified health data for claims

Registration Flow (4 Steps)

- Step 1 — Organization Type: Select from Corporate / Hospital / Gym / School / etc.
- Step 2 — Organization Details: Name, contact email, phone, location, initial seat capacity
- Step 3 — Email Verification: 6-digit OTP sent to contact email (dev mode shows OTP in response)
- Step 4 — Admin Account: Set primary admin name + secure password (bcrypt 12 rounds)
- On Success: Unique Org Code generated (e.g., ORG-XXXX) for employee enrollment

Login Flow (2FA)

- Step 1: Enter email + password !' credentials verified
- Step 2: 6-digit OTP sent to admin email !' verify OTP
- Step 3: JWT business token issued !' access to portal dashboard
- Sessions are JWT-based (BUSINESS_JWT_SECRET), separate from user tokens

Portal Pages & Routes

Route	Page Name	Functionality
/	Landing Page	Public marketing page with Get Started / Login CTAs
/login	Login	2FA email + OTP authentication
/register	Register	4-step organization onboarding wizard
/dashboard	Dashboard	KPIs, enrollment code, aggregate health metrics
/members	Members	Searchable directory, health profiles, plan management
/analytics	Analytics	Charts: demographics, trends, risk distribution
/communications	Communications	Send announcements/alerts to all members
/codes	Enrollment Codes	Multi-code management with seat limits + expiry
/billing	Billing	Seat subscription, GST invoices, Razorpay payments
/settings	Settings	Admin profile + organization details

05 Business Portal — Continued

Dashboard Analytics (Privacy-Safe)

- Total Members — Count of enrolled users with seat utilization %
- Active Users (7-day) — Members who logged health data in last 7 days
- Average Health Score — Platform-wide mean score (0-100) across all members
- Health Distribution — Pie chart: % Healthy vs At-Risk vs Inactive
- Daily Active Trend — 14-day bar chart of daily user engagement
- Aggregate Metrics — Nutrition, Hydration, Exercise, Medicine compliance averages
- Demographics — Gender, Age group, and Plan distribution charts
- Join Trends — Monthly member growth chart

Member Management (CRM)

- Searchable Directory — Search by name, phone, or 12-digit Aorane ID
- Member Health Profile — Blood group, BMI, gender, 5-day health score trend
- Plan Management — View and modify individual member subscription tier
- Remove Member — Admin can remove an employee from the organization
- Data Export — CSV export of member list for HR or compliance
- Org Code Enrollment — Share unique code for self-enrollment via mobile app

Communication Engine

- Announcement — General updates pushed to all member mobile apps
- Health Alert — Urgent wellness alerts (e.g., "Reminder: Annual Health Checkup")
- Reminder — Scheduled wellness nudges (water intake, step goals, etc.)
- Info Update — Share health tips or company wellness policy updates
- Delivery — Messages appear as in-app notifications in AORANE mobile app

Pricing Plans

Plan	Price	Members	Key Features
Starter	Free	Up to 10	Basic dashboard, standard health logging
Max	¹ 199/user/mo	Up to 500	Full analytics, AI scorecard, blood bank, priority support
Pro	¹ 249/user/mo	Unlimited	Custom branding, API access, dedicated account manager

Billing supports Monthly/Yearly cycles (15% discount on annual). Fully integrated with Razorpay. Automated GST invoice generation (CGST/SGST/IGST) for Indian businesses.

06 Admin Console — Internal Management

Overview

A powerful internal superadmin panel for full-spectrum platform governance. Dark-themed UI (Tailwind + Recharts). Access restricted to authorized personnel only via email + bcrypt password authentication. Every action logged in immutable audit logs.

Authentication

- Login URL: /admin-panel/
- Credentials: Email + password (SHA-256 legacy !' auto-upgraded to bcrypt)
- Default Admin: admin@zorane.com / admin123 (must change in production)
- Brute Force Protection: Max 10 attempts per email per 15 minutes
- JWT Token: Stored in localStorage as ap_token
- Role: SuperAdmin (full access to all modules)

Admin Modules — Complete List

Module	Section	Key Capabilities
Dashboard	Core	Platform KPIs: total users, orgs, revenue, active sessions
Users	Core	Search, ban/unban, activate/deactivate, plan override
Organizations	Core	B2B org list, seat allocations, org admin details
Revenue & Business	Finance	MRR, net profit/loss, Razorpay transaction logs
Plan Pricing	Finance	Configure Free/Pro/Max/Family plan prices and features
Subscriptions	Finance	Active license management, expiry tracking
Analytics	Finance	Platform-wide growth charts, AI-driven insights
Platform Costs	Finance	Infrastructure spend tracking vs revenue (breakeven)
Ads Manager	Content	Promotional banner management for mobile app
AI Config	Content	Per-feature AI model assignment + system prompt editing
Feature Flags	Content	Real-time enable/disable of any app feature
Food Database	Content	Curated nutritional database management
Promo Codes	Content	Discount code creation, tracking, expiry
Announcements	Content	Platform-wide user notifications
Blood Emergency	Emergency	Request verification, status control, moderation
Languages	System	Multi-language translation management
Audit Logs	System	Immutable record of all admin actions with timestamps
Upcoming Features	System	Roadmap/Coming Soon section management

06 Admin Console — Deep Features

User Management

- Search by Aorane ID (12-digit) or phone/name
- View blood group, BMI, city, age for support context
- Ban/Unban user (reason logged in audit trail)
- Activate/Deactivate account
- Manual plan override — upgrade any user to Pro/Max without payment

Financial Intelligence

- MRR (Monthly Recurring Revenue) — Real-time from active subscriptions
- Net Profit = Revenue - Razorpay fees (2%) - Fixed operational costs (\$87/mo)
- Breakeven Analysis — Target: X users at ₹199/mo needed to reach profitability
- Transaction Log — Every Razorpay payment ID with status (success/failed/pending)
- Platform Cost Tracker — Render hosting, Supabase DB, AI API costs vs revenue

AI Configuration System (Unique Feature)

Admins can assign different AI models to different features without any code changes:

Feature	Configurable Options
food_ai (text)	NVIDIA LLaMA 3.3 70B, Gemini 2.0 Flash, GPT-4o mini
food_scan (image)	Gemini 2.0 Flash Vision, GPT-4o Vision
health_prediction	DeepSeek V3, LLaMA 3.3 70B, Gemini Pro
smart_scan	Gemini 2.0 Flash Vision (OCR for reports)
meal_planner	Gemini Pro, LLaMA 3.3 70B
stress_insight	Any configured model

System prompts for each AI feature can also be edited directly from the Admin UI, enabling prompt optimization without developer intervention.

Feature Flags System

Real-time on/off toggles for any platform module. Changes propagate to mobile app without requiring an app update:

- smart_scan — Enable/disable AI medical report scanning
- meal_planner — Enable/disable AI diet plan generation
- health_suggestions — Enable/disable daily AI health tips
- wearable_sync — Enable/disable Google Fit integration
- blood_bank — Enable/disable blood donation network
- Any new feature can be added as a flag before full release

Blood Emergency Moderation

- View all active blood requests across India in real-time
- Verify hospital name, units needed, doctor contact
- Mark requests as Fulfilled (when blood found) or Cancelled
- Flag suspicious or duplicate requests to prevent platform abuse
- Platform safety mechanism — prevents fake emergency requests

07 Backend API Server

Overview

A centralized Express.js + TypeScript REST API serving all four client applications. Runs on Render (Singapore region). Uses Drizzle ORM for type-safe PostgreSQL queries. Handles authentication, business logic, AI orchestration, and payment processing.

API Modules — Complete Endpoint Reference

Module	Base Path	Key Endpoints
Authentication	/auth	OTP send/verify, Firebase login, Google OAuth, PIN, email OTP, refresh token
Users	/users	Profile CRUD, onboarding, medical conditions, health goals, account, Avatar ID
Food	/food	Log meal, AI scan, search DB, favorites, weather suggestions, summary by date
Health Tracking	/health	Exercise log/calculate, water log, health score compute, weekly activity
Medicine	/medicine	Schedule CRUD, dose logging (taken/missed), compliance tracking
Stress	/stress	Multi-mode log (five-pillar/clinical/mood), today summary, weekly baseline
Blood	/blood	Donor register/search, emergency request, GPS near-me, donation confirm
Family	/family	Create group, join with code, member health overview
Period	/period	Cycle log, symptom tracking, AI cycle prediction
AI	/ai	Diet plan, meal swap, smart scan (food/medical/medicine)
Intelligence	/health/intelligence	Disease risk prediction, weekly diet chart generation
Business	/business	Org register/login, member management, enrollment codes, billing
Admin	/admin	Platform overview, user moderation, feature flags, revenue, AI config
Payments	/payment	Razorpay order/verify, subscription management, webhook handler
Sessions	/sessions	DAU/MAU tracking, session heartbeat
Suggestions	/suggestions	Daily AI health suggestions, notification settings
Wearable	/wearable	Google Fit token exchange, data sync
Emergency	/accident-emergency	SOS logging, emergency contact management
Notifications	/notifications	Settings management (water/medicine/exercise reminders)

08 Database Schema (PostgreSQL + Supabase)

Database

Hosted on Supabase (PostgreSQL 15). Managed via Drizzle ORM with TypeScript type-safe queries. Shared across all backend modules. 115 startup migrations run on each deploy.

Core User Tables

- users — Auth: phone, email, created_at, role
- user_profiles — Physical data: height, weight, BMI, blood group, gender, work_profile, sleep_hours_avg
- user_preferences — App settings: language, dark mode, goals
- user_privacy_settings — Data sharing controls per category

Health Tracking Tables

Table	Purpose
food_logs	Meal entries with calories, macros, meal type, food name, AI-generated flag
exercise_logs	Workout sessions with type, duration, intensity, MET-based calories
water_logs	Daily water intake per glass entry with timestamp
medicine_schedules	Prescription plans: medicine name, dosage, timing, frequency
medicine_logs	Dose compliance: taken/missed/skipped with timestamp
stress_logs	Multi-type stress entries with score, mood, type, five-pillar data
period_logs	Menstrual cycle data: start date, flow, symptoms
daily_health_scores	Computed daily score (0-100) with pillar breakdown
health_predictions	30-day disease risk predictions per user
weekly_diet_charts	AI-generated 7-day meal plans per user

Business & Platform Tables

- organizations — Org details, seat count, enrollment codes
- org_admins — Business portal admin credentials
- org_members — Many-to-many: users enrolled in organizations
- enrollment_codes — Multi-code management with seat limits + expiry
- subscriptions — Active user plans with Razorpay subscription IDs
- payments — Transaction log for all Razorpay payments
- plan_pricing — Configurable plan features and prices
- feature_flags — Real-time feature on/off toggles
- food_items — Curated nutritional database (admin-managed)
- food_scan_cache — AI scan results cached to reduce API costs
- admin_audit_logs — Immutable admin action log
- ai_config — Per-feature AI model + system prompt configuration
- blood_donors / blood_emergency_requests / blood_donations
- emergency_contacts — SOS contact list per user
- app_sessions — DAU/MAU session tracking

09 Complete Tech Stack & Integrations

Frontend Technologies

Technology	Version	Used In
React	18	Landing, Business Portal, Admin Panel
Expo + React Native	54 / 0.81	Mobile App (iOS/Android/Web)
TypeScript	5.x	All packages
Tailwind CSS	4.x	Landing, Business Portal, Admin Panel
Vite	7.x	Landing, Business Portal, Admin Panel build tool
Expo Router	6.x	Mobile app navigation
Framer Motion	12.x	Landing page animations
Recharts	2.x	Charts in Landing, Admin, Business Portal
Radix UI	latest	Accessible UI primitives
Lucide React	latest	Icon system across all web apps
Ionicons	5.x	Icon system in mobile app

Backend Technologies

Technology	Purpose
Express.js + TypeScript	Core API server framework
Drizzle ORM	Type-safe PostgreSQL queries + migrations
PostgreSQL (Supabase)	Primary database + connection pooling
pnpm workspaces	Monorepo package management
ESBuild	Fast TypeScript bundling for API server
Pino Logger	Structured JSON logging
bcrypt	Password hashing (12 rounds)
jsonwebtoken	JWT generation and verification

External Service Integrations

Service	Provider	Used For
AI (Text)	NVIDIA API (LLaMA 3.3 70B)	Food nutrition analysis, health insights
AI (Vision)	Google Gemini 2.0 Flash	Food photo scan, medical report OCR
Payments	Razorpay	Subscriptions, business billing, GST invoices
Auth	Firebase	Phone OTP, Google OAuth, ID token verify
Email	Resend	OTP delivery, health reports, welcome emails
SMS	Fast2SMS + Twilio	OTP delivery via SMS fallback chain
Wearable	Google Fit API	Steps, heart rate, sleep, SpO2 sync
Weather	OpenWeather API	Food suggestions based on local weather
WhatsApp	Aisensy (planned)	WhatsApp bot (Coming Soon)

10 Subscription & Monetization

Individual Plans (Mobile App)

Plan	Price	Key Features
Free	₹0/month	Basic tracking (food, water, exercise, medicine), 5 AI scans/month
Pro	₹149/month	Unlimited AI scans, health predictions, PDF reports, family add-on
Max	₹249/month	Everything in Pro + diet chart, wearable sync, priority AI, premium support
Family	₹399/month	Max features for up to 4 family members under one account

Business Plans (Organization Portal)

Plan	Price	Members	Key Features
Starter	Free	Up to 10	Basic dashboard, standard health logging
Max	₹199/user/mo	Up to 500	Full analytics, AI scorecard, blood bank, priority support
Pro	₹249/user/mo	Unlimited	Custom branding, API access, dedicated account manager

Revenue Model

- Primary: SaaS subscriptions (individual + business seat-based)
- Gateway: Razorpay — 2% per transaction fee (accounted in P&L)
- Annual discount: 15% off for yearly billing (improves cash flow)
- Promo codes: Admin-managed discount codes for acquisition campaigns
- Breakeven target: ~500 Pro subscribers (₹149/mo) or ~340 Max subscribers

Payment Infrastructure

- Razorpay Orders API — Creates order before payment
- Razorpay Subscriptions — Recurring billing for monthly/yearly plans
- Webhook Handler — /payment/webhook verifies payment signature and activates plan
- GST Invoicing — Automated CGST/SGST/IGST calculation for Indian compliance
- Business Billing — Bulk seat purchase with Razorpay Orders for org admins

11 Security & Compliance

Data Security Measures

Layer	Implementation
Encryption in Transit	HTTPS/TLS 1.3 enforced across all endpoints
Password Hashing	bcrypt (12 rounds) for all user and admin passwords
JWT Tokens	Separate secrets for users / admins / business portals
OTP Verification	All sensitive actions (login, blood request, donor register) require OTP
PIN Lock	App-level 4-digit PIN re-authentication on return to app
Rate Limiting	OTP: 1/hour per phone. Admin login: 10 attempts/15min
Brute Force	IP + email based lockout for admin panel
Audit Logs	Every admin action recorded with timestamp + actor + target
Webhook Signature	Razorpay webhook signature verified before processing payment
Feature Gating	Per-feature requireFeature() middleware + plan check

Privacy & Regulatory Compliance

- DPDPA Act 2023 (India) — Individual health data not exposed to org admins; only aggregate metrics
- HIPAA-Ready Architecture — Designed for healthcare data handling standards
- Data Minimization — Only required data collected during onboarding
- User Privacy Controls — Granular per-category data sharing toggles in settings
- Data Export — Users can export their own health data (right to data portability)
- Right to Erasure — Account deletion removes all personal health data
- Firebase Auth — Phone verification handled by Google (no plaintext phone storage)

12 Deployment Architecture

Production Deployment

Component	Platform	URL / Domain
API Server	Render (Singapore)	https://aorane.onrender.com
Landing Page	Vercel	https://aorane.com
Business Portal	Vercel (separate project)	TBD — business.aorane.com
Admin Panel	Vercel (separate project)	TBD — admin.aorane.com
Mobile App (Web)	Expo Hosting / Replit	Web preview via Expo domain
Mobile App (APK)	Codemagic CI/CD	Play Store — in.aorane.app
Database	Supabase (PostgreSQL)	Supabase cloud instance

CI/CD & Auto-Deploy

- Source Control: GitHub — github.com/Mangesh-a-verma/Health-Data-Hub
- Render: Auto-deploy on push to main branch (render.yaml in repo root)
- Vercel: Auto-deploy on push to main (vercel.json in each artifact folder)
- Codemagic: APK build on trigger — uses codemagic.yaml + eas.json
- Commit Author: Must use GitHub-linked email for Vercel to recognize deployer
- API Build: `pnpm --filter @workspace/api-server build !' esbuild bundle`
- Startup: 115 DB migrations run on each API server start

Environment Variables — Production Secrets

Variable	Service	Description
DATABASE_URL	Render	Supabase PostgreSQL connection string
JWT_SECRET	Render	User authentication token signing key
JWT_REFRESH_SECRET	Render	Refresh token signing key
ADMIN_JWT_SECRET	Render	Admin panel token signing key
BUSINESS_JWT_SECRET	Render	Business portal token signing key
NVIDIA_API_KEY	Render	NVIDIA AI API access for LLaMA model
RESEND_API_KEY	Render	Email delivery for OTPs and reports
RAZORPAY_KEY_ID + SECRET	Render	Payment processing
RAZORPAY_WEBHOOK_SECRET	Render	Webhook signature verification
GOOGLE_FIT_CLIENT_ID/SECRET	Render	Google Fit OAuth integration
EXPO_PUBLIC_FIREBASE_API_KEY	Mobile	Firebase config (safe to be public)

AORANE

•Y "Ù " 8"y9'BÂ 9'éG"i> 'YG ")?•đ

Document prepared by AORANE AI Engineering Team

© 2025 AORANE Technologies Pvt. Ltd. — CONFIDENTIAL & PROPRIETARY

This document contains proprietary information about the AORANE platform architecture and product strategy.

Unauthorized distribution is strictly prohibited.

